

ARIAL ENGINEERING SERVICES

Technology & Security Questionnaire — Vendor Response (REVISED v3)

VENDOR RESPONSE DOCUMENT — AES-TECH-2026-001 REVISED v3

Updated 09 June 2026 to reflect client decisions confirmed on the 9 June call

Document Ref	AES-TECH-2026-001 REVISED v3
Date of Vendor Response	09 June 2026
Supersedes	AES-TECH-2026-001 REVISED v2 (dated 06 June 2026)
Related Quotation	AES-Q-2026-003 (Professional Full-Suite) + AES-SOW-2026-003 v3
Completed By	Mahender Vengala & Hitansu Parichha Arial Engineering Solution
Response Covers	Professional Full-Suite — ₹6,79,500 (unchanged)

What changed in this Revision v3

- Hosting model: Arial-provided dedicated server (NOT cloud). All 4 environments on one server via Docker.
- Authentication: Google Sign-In + Email/Password. SMS OTP retained ONLY for payment confirmation and job-closing verification.
- WhatsApp scope expanded: full in-chat service flow included alongside booking/ETA/job-complete alerts.
- Email provider switched from SendGrid to Amazon SES (~\$12/year at our volume).
- Razorpay terms clarified: UPI / domestic debit cards 0%, credit cards / netbanking 2% + 18% GST on platform fee only.
- Sprint methodology: 2-week sprints with a shared Excel requirement tracker.
- Server provisioning timing: required from Arial by end of Week 6 (~Day 45).
- Currency convention: USD-priced services shown in USD (\$99/year Apple, \$25 Google Play, ~\$12/year email). INR-native services in ₹.

This document is our formal written response to every question in the revised questionnaire AES-TECH-2026-001 REVISED. All commitments here form part of the agreed project terms and will be incorporated as Annexure A of the signed SOW v3.

Section 1 — Product & Solution Overview

What is being built, who it is for, and how it fits together

1.1 What is the name of the platform being built?

The final commercial brand name will be confirmed by Ariel Engineering Services during the Discovery phase (Sprint 1). Whichever name Ariel selects will be embedded immediately across the Web Portal, Android app, iOS app, App Store / Play Store listings, login screens and all documentation. The platform is referenced as "AES Service Application" in the quotation and SOW as a working placeholder only.

1.2 Describe the platform in 3–4 sentences: what it does, who uses it, and the core problem it solves for Ariel Engineering.

The AES Service Application is a unified digital service management platform that enables Ariel Engineering's customers to book AC installations, raise service complaints, view AMC contracts, approve quotes, make online payments and track engineers in real time — delivered across a responsive Web Portal, a native Android app and a native iOS app.

Internally, it provides Ops Managers, CRM Agents, Site Engineers, Service Managers and Admins a single workbench to triage tickets, dispatch engineers, escalate SLA breaches, manage parts and quote approvals, and monitor live revenue.

The platform replaces phone-call-driven, spreadsheet-based service operations with a fully audited, real-time system purpose-built for Ariel's HNI/UHNI clientele.

The core problem it solves is the loss of revenue and customer trust caused by missed SLAs, manual dispatch and the absence of customer visibility into ongoing service work.

1.3 Confirm which modules are included in Phase 1 scope:

- ✓ **Customer Portal (Web + Mobile)**
- ✓ **Service / Complaint Ticketing**
- ✓ **CRM Agent Workbench**
- ✓ **Ops Manager Dashboard**
- ✓ **Admin Earnings & Analytics**
- ✓ **SLA Tracking & Escalation**
- ✓ **Live Engineer Tracking**
- ✓ **Quote & Parts Approval**
- ✓ **AMC Contract Module**
- ✓ **WhatsApp Integration (full — including in-chat service flow)**
- ✓ **Payment Gateway (Razorpay)**
- ✓ **Employee / Technician Management**

Additional modules included in Professional scope: Product Catalogue, Error-Code Diagnostic Guide, Notification Centre, Complete Audit Log & Activity Trail, Multi-Property Address Book per customer.

1.4 What is explicitly OUT OF SCOPE for Phase 1?

The following items are NOT included in the contract value of ₹6,79,500. They are third-party costs payable directly by Ariel Engineering Services. Updated to reflect the 9 June decisions:

Service / Item	Approximate Cost	Notes
Dedicated server (all environments)	₹2,000 – ₹12,000 / month	Ariel chooses provider (VPS / managed / on-premise). NOT cloud — per 9 June decision.
Apple Developer Program	\$99 / year (USD)	Annual renewal required for App Store presence
Google Play Developer Account	\$25 one-time (USD)	One-time lifetime fee
Google Maps Routes API (engineer ETA)	\$5 – \$20 / month (~\$60 – \$240 / year)	Maps SDK mobile is FREE (India SKU). Only ETA routing is paid.

Service / Item	Approximate Cost	Notes
WhatsApp Business API (Gupshup / Meta)	~₹0.20 per utility msg + platform fee	Includes full in-chat service flow. ~₹300–₹500/month at Arial's expected volume.
Amazon SES (transactional email)	\$0.10 per 1,000 emails (~\$12 / year)	OTPs, receipts, AMC reminders. Replaces SendGrid.
SMS via MSG91 (DLT-registered)	₹0.20 per SMS	Used ONLY for payment OTP + job-close OTP. ~₹40/month at 100 transactions.
DLT registration (TRAI)	₹15,000 – ₹25,000 one-time	Mandatory for any SMS in India. Required before Go-Live.
Razorpay transaction fees	Variable — see Section 8.1	0% on UPI / debit; 2% + 18% GST on credit cards & netbanking
GitHub Team plan (Arial org)	~\$96 / year (2 dev seats)	Required for private repos in Arial's GitHub organisation
Domain name registration / renewal	₹800 – ₹2,000 / year	Cheapest at Cloudflare Registrar; if Arial already owns the domain, ₹0 additional
SSL certificate	FREE	Let's Encrypt — auto-renewed every 90 days by the deployment script

Other items outside scope: hardware for Arial staff (laptops, tablets, smartphones); legacy data migration (priced separately once data volume confirmed); features not listed in Sections 1.3 / 1.5; promotional / marketing WhatsApp campaigns (utility messaging is included).

1.5 Confirm all third-party integrations planned for Phase 1:

- ✓ **WhatsApp Business API (Gupshup OR Meta Cloud API) — full integration incl. in-chat service flow**
- ✓ **SMS Gateway (MSG91, DLT-registered) — ONLY for payment OTP and job-close verification**
- ✓ **Payment Gateway (Razorpay)**
- ✓ **Google Maps Platform — Maps SDK (Android), MapKit (iOS), Maps JS API (Web), Routes API for engineer ETA**
- ✓ **Firebase Cloud Messaging (FCM) — Android push notifications (FREE)**
- ✓ **Apple Push Notification service (APNs) — iOS push notifications (FREE)**
- ✓ **Amazon SES — transactional email (OTPs, receipts, reminders, reports)**
- ✓ **Google Sign-In (OAuth via Firebase Authentication / Spark plan — FREE) — primary login**

Storage: file uploads (engineer photos, customer documents, AMC PDFs, invoice attachments) stored on the dedicated server's data disk. NOT cloud storage. Encrypted at rest by the filesystem.

Section 2 — Technology Stack

A clear record of every technology powering this platform

2.1 Complete technology stack

Layer	Technology / Detail
Web Front-End	Next.js 16 + React 19 (App Router, Turbopack), Framer Motion, Lucide React
Backend Language & Framework	Java 21 + Spring Boot 3.3.5 (spring-web, spring-data-jpa, spring-security, spring-websocket, spring-validation)
Mobile — Android	Kotlin + Jetpack Compose, min SDK 26 (Android 8.0), target SDK 34
Mobile — iOS	Swift 5.9 + SwiftUI, deployment target iOS 15+
Primary Database	PostgreSQL 15 — running in Docker on Ariel's dedicated server (NOT Cloud SQL)
Cache / Secondary store	Redis 7 — running in Docker on Ariel's server (session cache, rate-limit, WebSocket presence)
API Architecture	REST (JSON over HTTPS), documented via OpenAPI 3 / Swagger UI
Real-Time / WebSocket	Spring WebSocket + STOMP; client uses @stomp/stompjs + sockjs-client
Authentication	PRIMARY: Google Sign-In (OAuth via Firebase Auth Spark plan — FREE) + Email/Password (BCrypt-hashed). JWT (jjwt 0.12.3) for session tokens (access 15 min + refresh 30 days). Admin MFA via Google Authenticator (TOTP). SMS OTP retained ONLY for payment confirmation and job-close verification.
Version Control	GitHub — private repos in Ariel Engineering Services' organisation (Ariel creates org; dev team added as Admin)
CI / CD	GitHub Actions → Docker build → deploy to Ariel's server via SSH. Cloud Build NOT used (no GCP).
Containerisation	Docker (multi-stage builds) + Docker Compose for orchestration on Ariel's server
Android Push	Firebase Cloud Messaging (FCM) — FREE
iOS Push	Apple Push Notification service (APNs) — FREE
SMS / WhatsApp	MSG91 (SMS, DLT-registered) — limited use; Gupshup OR Meta Cloud API direct (WhatsApp Business)
Payment Gateway	Razorpay (Checkout + Webhooks + Refund API)
Email Service	Amazon SES (transactional only; pay-as-you-go \$0.10 per 1,000 emails)
File / Media Storage	Dedicated server's data disk (encrypted filesystem) with signed-URL access from the API
Hosting	Ariel-provided dedicated server (Linux, Ubuntu 22.04 LTS recommended). All 4 environments (Dev / QA / Staging / Production) on the same server, isolated via Docker.
Region / Data residency	India (per server provider). All customer data stays on Ariel's server inside India.

Note on mobile framework: native (Kotlin + Swift/SwiftUI) is the recommended implementation. Cross-platform (React Native) remains an option to be confirmed in writing during Discovery (Week 1) based on agreed performance, time-to-market and long-term maintenance trade-offs. Either path delivers the full 44-screen feature set.

2.2 Will the Git repository be in Arial Engineering's GitHub / GitLab account or the development team's?

☒ Repository in Arial Engineering's account from Day 1

- ☐ Starts in dev team's account, transferred at final delivery
- ☐ Shared — both parties have admin access
- ☐ To be arranged

Confirmed on 9 June call: Arial Engineering Services creates a private GitHub organisation under Arial's name (e.g. arial-engineering). Arial adds Mahender Vengala and Hitansu Parichha as Admins on four repositories: aes-frontend, aes-backend, aes-android, aes-ios. All commits push directly into Arial's organisation from Day 1. Full commit history, pull requests, issues and CI logs remain Arial's property at all times.

2.3 Are any proprietary frameworks or licensed libraries used that Arial would need to license separately after handover?

- ☐ Yes — described below
- ☒ **No — all open source or standard SDKs**
- ☐ Not yet determined

Every dependency is either open-source (Apache 2.0 / MIT: Spring Boot, Next.js, React, Framer Motion, Jetpack Compose, SwiftUI, jjwt, Lombok, SpringDoc) or a free-tier vendor SDK that Arial signs up for and pays directly (Razorpay, Firebase, Apple, MSG91, Gupshup, Google Maps, Amazon SES). No software licence is payable to the Service Provider. The Service Provider absorbs its own commercial productivity tools (IDE subscriptions, code-security scanners) out of the contract value — never passed through to Arial.

Section 3 — Infrastructure & Hosting

Where the platform lives and who controls it

3.1 Will the hosting be in Arial Engineering's name or the development team's?

☒ **Arial Engineering provides a dedicated server. Arial controls the server, billing and root access.**

☐ Development team's account — Arial given admin / owner access

☐ Development team's account — Arial has no direct account access

☐ To be arranged prior to go-live

Confirmed on 9 June call: Arial Engineering Services provisions one dedicated server (Linux, recommended Ubuntu 22.04 LTS). All four environments (Dev / QA / Staging / Production) run as isolated Docker containers on the same server. The Service Provider receives SSH access (via SSH keys) for Mahender Vengala and Hitansu Parichha throughout the build, UAT and 30-day warranty period. After warranty ends, Arial may downgrade or remove SSH access. The server, its billing, and all data remain entirely Arial's property throughout.

3.2 Confirm which environments will be provisioned:

Statement	Yes	No	Partial
Development environment (isolated Docker stack, for active coding)	✓		
Testing / QA environment (separate Docker stack, separate database)	✓		
Staging / UAT environment (mirrors production exactly — same Docker images)	✓		
Production environment (live, customer-facing)	✓		

All four environments run on Arial's single dedicated server, isolated via Docker networks. Each environment has its own PostgreSQL container, its own Redis container, its own backend container, its own port mapping. Staging and Production share machine specs; Dev and QA can be down outside business hours to save server load.

3.3 Estimated monthly infrastructure cost for the first 12 months (server + third-party services):

All costs below are paid directly by Arial to the respective providers. The Service Provider takes zero margin on any of these.

Item	Estimated Cost
Dedicated server — VPS option (DigitalOcean / Linode / Hetzner)	₹2,000 – ₹4,500 / month
Dedicated server — managed (CtrlS / ESDS Hyderabad)	₹6,000 – ₹12,000 / month
Apple Developer Program	\$99 / year (USD)
Google Play Developer Account	\$25 one-time (USD)
Google Maps Routes API (engineer ETA)	\$5 – \$20 / month (~\$60 – \$240 / year)
WhatsApp Business API (Gupshup utility messages)	~₹300 – ₹500 / month at expected volume
SMS via MSG91 (payment OTP + job-close OTP only)	~₹40 – ₹150 / month at 100–500 transactions
Amazon SES (transactional email)	~\$1 / month (~\$12 / year)

Item	Estimated Cost
Domain name renewal	₹800 – ₹2,000 / year
GitHub Team plan (Arial's GitHub organisation)	\$4 / user / month (~\$96 / year for 2 dev seats)
Total estimated Year 1 recurring (excluding Razorpay txn fees)	~₹36,000 – ₹1,80,000 / year + ~\$285 / year USD-priced items

How this compares with v2

v2 estimate was ₹1,40,000 – ₹2,10,000 / year because it assumed Google Cloud hosting at ₹8,500 – ₹12,000/month.

v3 estimate is dramatically lower because hosting moves to a single dedicated server (Arial's choice of provider). The biggest annual saving is ~₹1,00,000 in cloud charges that no longer apply.

The trade-off: HA, automated managed backups and zero-config scaling that come with cloud are replaced by the Service Provider's server runbook + Docker-Compose configuration + nightly backup scripts.

3.4 Who is responsible for server and infrastructure maintenance after go-live?

☒ **Development team — included in the optional Support Plan after Month 4 warranty**

☐ Arial Engineering's internal team

☐ Shared responsibility — defined in support agreement

☐ To be agreed

Month 4 warranty (included in contract): server monitoring, OS / Docker patching, application bug fixes, backup verification, uptime alerting — all at no cost. After Month 4: Arial may activate Basic Maintenance plan (₹12,000/month) for continued server + app care, or take over with internal / third-party DevOps. Full server runbook and Docker-Compose configuration provided at Go-Live.

Section 4 — Delivery, Milestones & Accountability

Specific commitments on timeline, quality gates and escalation

4.1 Proposed project start date:

Within 2 business days of receipt of the M1 advance payment and a countersigned SOW v3. The 13-week development calendar is locked in Week 0 (kick-off) and shared with Arial. A rolling sprint plan ensures new requirements raised during Discovery adjust the schedule without confusion.

4.2 Milestone schedule — 5 equal milestones of 20% each (₹1,35,900)

Confirmed on 9 June call. 5 equal milestones replace the original 30/20/30/20 split. Each milestone is sized on a 2-week sprint cycle (Friday-to-Friday). Acceptance criteria are documented in detail in SOW v3 Section 8 (Definition of Development Sign-off).

Milestone	Trigger / Cycle	Deliverable & Sign-off Criteria
M1 — Kick-off	Week 0 (signing day)	Signed SOW; project plan, sprint calendar (8 sprints), shared Excel tracker URL, communication groups set up
M2 — Discovery Complete	End of Sprint 1 (Week 2)	BRD, workflow maps (30+ scenarios), role-permission matrix, technical architecture document, integration plan — all signed off by Arial PM
M3 — UI/UX Approved	End of Sprint 2-3 (Week 5)	Hi-fi mockups for all 44 screens, interactive Figma prototype, design system handoff — Arial approves in writing
M4 — Dev Complete / Demo	End of Sprint 5-6 (Week 10–11)	Web + Android + iOS deployed to Staging on Arial's server; all 24 REST modules live; Razorpay test transaction completed; WhatsApp test message sent; map tracking working; demo sign-off signed; zero Critical/Major bugs open
M5 — UAT & Go-Live	End of Sprint 7 (Week 13)	UAT executed by Arial; zero Critical/Major bugs; signed UAT form; Web Portal live on Arial's domain; Android live on Play Store; iOS live on App Store; training sessions 1–3 delivered

Each milestone payment is ₹1,35,900 (20%). Calendar dates confirmed in Week 0 once M1 is received. All milestones land on a Friday (14-day Sprint cycles) so Arial can review over weekends.

4.3 Process if a milestone is delayed:

- ☒ Written notification to Arial at least 5 working days before expected delay
- ☒ Revised timeline agreed in writing with no additional cost to Arial
- ☒ Escalation to senior team lead immediately
- ☐ No formal delay process currently defined

4.4 Willingness to agree to a written milestone commitment with notification obligation:

- ☒ Yes — happy to include formal written commitments
- ☒ Yes — willing to agree notification SLA
- ☐ Open to discussion
- ☐ Prefer best-efforts basis only

Both the milestone schedule and the delay-notification SLA are documented in SOW v3 Section 8 and treated as Annexure A to the signed SOW.

4.5 How will project progress be communicated during the build?

- ☒ Weekly written status update (every Friday by 6 pm IST)
- ☒ Shared Excel sheet for requirement tracking (per 9 June decision — replaces Linear/Notion)
- ☒ 2-week sprint cycle — sprint planning Monday, demo on second Friday
- ☒ Bi-weekly demo call (end of each sprint, 30 min on Zoom / Google Meet)
- ☒ WhatsApp group for urgent items and critical blockers only
- ☒ On-demand — Arial can request a status check or call at any time

Sprint structure: Monday planning (60 min), Wednesday mid-sprint check-in (15 min stand-up via WhatsApp), Friday of Week 2 sprint demo (30 min Zoom). Shared Excel tracker columns: Story ID | Description | Owner | Sprint | Status | Acceptance Criteria | Sign-off date. Maintained on Google Drive / Microsoft 365.

4.6 Single point of contact from the development team:

Full Name	Mahender Vengala
Role / Designation	Project Lead & Delivery Manager
Mobile / WhatsApp	+91 79894 78549
Email	vengalamahender83@gmail.com
Available Hours	Mon–Sat 09:30–19:30 IST; urgent items 24×7 on WhatsApp during build + warranty
Secondary contact	Hitansu Parichha — Lead Engineer (full-stack + DevOps + App Store/Play Store submissions)

Section 5 — Security, Data Protection & Ownership

How customer and business data is protected and who owns it

5.1 Authentication & Access Controls

Statement	Yes	No	Partial
Google Sign-In (OAuth) as primary login on all 3 platforms	✓		
Email + Password (BCrypt-hashed) as alternative primary login	✓		
SMS OTP — for payment confirmation step only	✓		
SMS OTP — for job-closing verification at customer end only	✓		
SMS OTP at login time		✓	
JWT session tokens (access 15 min + refresh 30 days)	✓		
Role-Based Access Control — 6 roles	✓		
Admin MFA via Google Authenticator (TOTP) — mandatory, not optional	✓		
Brute-force / rate-limit protection on all login endpoints	✓		
Session invalidation on logout	✓		

Per the 9 June decision: SMS OTP is intentionally removed from login flows because users find it slow and the SMS cost is significant. Google Sign-In + Email/Password give users a faster, more familiar experience. SMS OTP is retained ONLY at the two highest-value touchpoints: (a) confirming a payment before it is captured, and (b) the customer's job-closing verification (so the engineer cannot mark the job complete without the customer's consent).

5.1 (cont.) Encryption & Data Security

Statement	Yes	No	Partial
All API communication via HTTPS / TLS 1.2 or higher	✓		
Database data encrypted at rest (filesystem encryption on data disk)	✓		
Uploaded files stored on encrypted server data disk	✓		
Secrets in .env files (chmod 600) — never hardcoded in source	✓		
Payment data handled via Razorpay only — never stored on the server	✓		
Customer PII (phone, address) not logged in plain text	✓		

5.1 (cont.) Data Ownership

Statement	Yes	No	Partial
All data on the server belongs entirely to Ariah Engineering Services	✓		

Statement	Yes	No	Partial
Development team has no right to access, use, or share Arial's customer data	✓		
On contract termination, all data remains on Arial's server (Arial controls it)	✓		
No customer data stored outside India	✓		

5.1 (cont.) Backup & Recovery

Backup frequency	Daily automated PostgreSQL pg_dump + nightly file backup. Backup script delivered as part of server runbook.
Backup location	Local backup directory on server's data disk PLUS off-site copy via server provider's snapshot/backup service (e.g. DigitalOcean Spaces, CtrlS daily backup). Configuration finalised during Sprint 4.
Backup retention	7 daily + 4 weekly + 12 monthly rolling snapshots
Recovery Time Objective (RTO)	≤ 4 hours to restore full production from latest snapshot
Recovery Point Objective (RPO)	≤ 24 hours data loss (driven by daily backup cadence; configurable to hourly if Arial later requests)
Backup testing	First restore drill during Week 14 (UAT week); repeated quarterly post-launch

5.2 Logging and monitoring in place from Day 1 of production launch:

- ✓ Application error and exception logging (Logback → log files on server)
- ✓ API access logging with timestamps and user IDs
- ✓ Failed login attempt logging — alert after 5 consecutive failed attempts from same IP
- ✓ Server / infrastructure monitoring with alerts (Netdata or Prometheus + Grafana on the same server)
- ✓ Uptime monitoring with automated notification (UptimeRobot or BetterStack — free tier)
- ☐ No formal logging at launch

5.3 If a security incident or data breach occurs, what is the response process?

- (1) Immediate isolation of the affected component (firewall rule / service shutdown).
- (2) Written notification to Arial PM and Owner within 24 hours via email and WhatsApp — including impact assessment and scope of exposure.
- (3) Root-cause analysis report delivered within 7 working days.
- (4) Remediation deployed and independently verified before re-enabling the affected service.
- (5) Post-incident review meeting with Arial leadership.

For confirmed personal-data breaches, Arial is fully supported in disclosure to affected users in line with the Indian DPDP Act 2023. A written Incident Response Plan is delivered as part of the Phase 1 documentation package.

Section 6 — Testing, QA & Acceptance Criteria

How quality is verified before Arial signs off

6.1 Confirm the testing approach for this project:

- ☒ **Unit testing — all critical backend API endpoints**
- ☒ **Integration testing — Web, Android, iOS with shared backend on Staging server**
- ☒ **Cross-device testing — min 3 Android + 2 iOS devices**
- ☒ **Cross-browser testing — Chrome, Safari, Firefox (latest 2 major versions)**
- ☒ **End-to-end testing for all 6 user roles**
- ☒ **Regression testing before each production release**
- ☒ **Performance / load testing before go-live (target: 200 concurrent users, p95 < 500ms)**
- ☒ **UAT session with Arial's team in Sprint 6**

Devices: Pixel 6, Samsung S22, OnePlus 11 (Android) + iPhone 12, iPhone 15 (iOS). E2E: Manual test cases for all 6 roles + Playwright scripts for critical flows (booking, payment, SLA escalation, engineer dispatch, AMC reminders).

6.2 UAT terms — rounds of client feedback and scope:

- ☐ 1 round of UAT — up to 20 bug fixes included
- ☒ **2 rounds of UAT — up to 30 bug fixes included**
- ☐ Unlimited UAT feedback within the project timeline
- ☐ UAT scope to be agreed separately

Arial receives credentials for the Staging environment (private URL + 6 test accounts, one per role). Bug reports filed in the shared Excel tracker with screenshots, expected vs actual behaviour and reproduction steps. Daily triage during UAT week: P1 (Critical) fixes within 24 hours, P2 (Major) within 48 hours. A signed UAT Sign-off form gates the Go-Live milestone.

6.3 Bug-resolution SLAs for post-go-live support:

Critical Bug — definition	Platform down; production data loss; payment / login / job-create completely non-functional; data exposed without authentication
Critical Bug SLA	Response within 1 hour — resolution within 4 working hours
Major Bug — definition	Single feature non-functional, workaround exists, no data loss (e.g. WhatsApp notifications not firing, AMC renewal emails not sending)
Major Bug SLA	Response within 4 hours — resolution within 24 working hours
Minor Bug — definition	Cosmetic / UI glitch, copy error, layout issue on specific device — no functional impact
Minor Bug SLA	Response within 1 business day — resolution within 5 business days

Section 7 — Post-Launch Support & Maintenance

Commitments after go-live

7.1 Post-launch warranty commitments:

Free warranty period	30 days post Go-Live (Month 4 of contract — already included in ₹6,79,500)
Warranty scope	Bug fixes for defects in delivered scope; server-side OS / Docker patching; backup script verification. Minor copy / text changes and UI tweaks included as a courtesy. New features require a written Change Request.
Critical bug response during warranty	Response within 1 hour — fix within 4 working hours
Minor bug response during warranty	Response within 1 business day — fix within 2 business days
Primary communication channel	WhatsApp group (Arial + dev team) for triage; email for formal sign-off; shared Excel tracker for ticket tracking
After warranty?	(a) Subscribe to a Support Plan (Section 7.2), OR (b) engage per-incident at ₹2,500/hour (min 2-hour block), OR (c) self-maintain using Arial's own / third-party developer

7.2 Optional post-launch plans — full terms:

Commitment	Basic ₹12K/mo	Standard ₹25K/mo	Premium ₹45K/mo
Critical bug SLA	Within 4 hours (email)	Within 2 hours (phone+email)	Within 1 hour (WhatsApp+phone+email)
Minor bug SLA	Within 2 business days	Within 1 business day	Within 4 business hours
Uptime SLA	99.0% monthly	99.5% monthly	99.9% monthly (credit-back if missed)
Server patching	Quarterly	Monthly	Continuous
New features?	No — patches + fixes only	Yes — up to 8 hrs/month	Yes — priority dev 10 hrs/month
Dedicated developer?	No (pooled team)	Partial — named account engineer	Yes — dedicated developer + backup
Monthly progress report?	No	Yes — by 5th of each month	Yes — emailed + monthly review call
Max support hours / month	6 hours	20 hours	40 hours

7.3 What is included in the training sessions? (upgraded from 3 to 8 sessions at no additional cost)

As part of the value-add offer, the standard 3 training sessions have been upgraded to 8 sessions x 1 hour each, spread across the first 4 months post-go-live.

#	Topic	Audience	Timing
1	Customer journey + Web Portal walkthrough	CRM team + Ops Manager	Week of go-live
2	Ops Manager triage workbench + engineer dispatch	Ops Manager + Senior CRM	Week of go-live

#	Topic	Audience	Timing
3	CRM Agent workbench — tickets, parts, quotes	CRM Agents	Week of go-live
4	Site Engineer mobile app + live tracking	All Site Engineers	Month 1
5	Service Manager — SLA escalation + parts approval	Service Manager	Month 1
6	Admin — Earnings Dashboard, Revenue Analytics, exports	Admin + Leadership	Month 2
7	AMC contract management deep-dive (renewals, reminders)	Ops + CRM	Month 3
8	Refresher + open Q&A workshop (any topic Arial picks)	All teams	Month 4

Each session conducted live on Zoom / Google Meet, recorded and shared with Arial. Each session includes a printable cheat-sheet PDF. All recordings remain Arial's property.

7.4 Documentation delivered at project completion:

- ✓ **User manual for each of the 6 roles**
- ✓ **API documentation / Postman collection**
- ✓ **Architecture and setup guide**
- ✓ **Database schema and Flyway migration scripts**
- ✓ **Server runbook (start / stop / backup / restore / restart procedures)**
- ✓ **Docker Compose configuration (commented) for all 4 environments**
- ✓ **Video walkthroughs (8 training-session recordings)**
- ✓ **In-app help text and tooltips**

Section 8 — Commercial Clarifications

Specific questions arising from AES-Q-2026-001, 002, 003 and AES-SOW-2026-003

8.1 Payment Gateway — Razorpay terms confirmed (per 9 June call)

☒ **Recommend Professional (₹6,79,500) — Razorpay included as standard**

☐ Recommend Standard — add payment gateway at ₹35,000 extra

☐ Recommend Basic (₹3,77,500) as Phase 1 MVP

Razorpay confirmed by Arial. Account creation is Arial's responsibility (free; requires PAN, GST and bank details). API keys and webhook secret to be shared with the dev team via a secure channel after activation.

Razorpay fee structure — confirmed on 9 June call

Payment instrument	Platform fee	GST
UPI (bank-to-bank)	0%	— (no fee)
Debit cards — domestic	0%	— (no fee)
RuPay credit cards via UPI	2.15%	+18% on the 2.15%
Visa / MasterCard credit cards	2%	+18% on the 2%
Netbanking	2%	+18% on the 2%
Wallets	2%	+18% on the 2%
EMI / Cardless EMI	3%	+18% on the 3%
International cards	3%	+18% on the 3%

GST calculation — answer to Pradeep's specific question of 9 June

Q: Does the 2% Razorpay fee apply on the base invoice amount or on the GST-inclusive total?

A: The 2% applies on the BASE INVOICE AMOUNT received from the customer — NOT on the GST-inclusive total.

And then 18% GST is applied ON THE PLATFORM FEE ITSELF (not on the full transaction value).

Worked example — customer pays ₹1,000 via a Visa credit card:

- Razorpay platform fee on base ₹1,000 = ₹20 (2%)
- GST on that ₹20 fee = ₹3.60 (18%)
- Total Razorpay deduction = ₹23.60
- Net amount settled to Arial's bank account = ₹976.40

On UPI / RuPay debit-card transactions, Razorpay does NOT charge any platform fee, so Arial receives 100% of those collections to its bank account.

8.2 WhatsApp Business API — full integration confirmed

☒ **Included in Professional package — confirmed via AES-Q-2026-003**

☐ Add to Standard package as add-on — additional charge TBD

☐ Use SMS fallback only in Phase 1 — WhatsApp in Phase 2

WhatsApp scope confirmed on 9 June: full integration including booking confirmations, engineer ETA alerts, job-complete notifications AND in-chat service flow (customer can converse with CRM through WhatsApp; messages routed into the platform inbox).

Provider: Gupshup OR Meta Cloud API direct — final choice confirmed during Discovery (Week 1). Recurring per-message cost (~₹0.20 per utility template) is billed by the provider directly to Arial. Service Provider takes zero margin.

8.3 AMC Module — full management confirmed

☒ **AMC module is included in Phase 1 scope — full management**

☐ AMC module requires additional scoping and pricing

☐ AMC module planned for Phase 2

AMC module fully included in Phase 1 — NOT a viewer only. Scope: contract creation/upload, expiry tracking with automated reminders at 90 / 60 / 30 / 15 / 7 days (WhatsApp + email + in-app), scheduled-visit auto-generation from AMC contracts, AMC vs paid-call differentiation on every service ticket, customer-facing AMC viewer on Web + Android + iOS, and renewal-quote generation workflow.

8.4 Backend technology — complete backend, frontend and server stack confirmed

Backend: Java 21 + Spring Boot 3.3.5 with REST APIs, PostgreSQL 15 in Docker on Arial's dedicated server, Flyway for schema migrations, JWT (jjwt 0.12.3) for authentication, Spring WebSocket + STOMP for real-time, SpringDoc OpenAPI for API documentation, MSG91 SDK for SMS, AWS SDK for Amazon SES.

Web Front-End: Next.js 16 + React 19 on Turbopack, Framer Motion, Lucide React, STOMP/SockJS client for real-time updates.

Mobile Apps: Native (Kotlin + Jetpack Compose for Android; Swift 5.9 + SwiftUI for iOS) recommended. Cross-platform (React Native) remains an option to be confirmed during Discovery based on agreed performance, time-to-market and long-term maintenance trade-offs. Either path delivers the full 44-screen feature set on both Play Store and App Store.

Server: dedicated Linux server provided by Arial (Ubuntu 22.04 LTS recommended). 4 vCPU / 8 GB RAM / 250 GB SSD minimum. All 4 environments isolated via Docker.

Stack consistency: Backend, frontend and mobile technology is identical across all three package tiers. Differences between tiers are feature scope only.

8.5 Apple Developer Account & Google Play Console — access required from Arial:

Apple Developer Program (\$99/year — USD) — Arial maintains the account in Arial's legal name:

1. App Store Connect — App Manager access for Hitansu Parichha (Lead Engineer) for the entire build, store-listing submission and 30-day post-launch warranty period. Arial downgrades or removes access after warranty ends. Hitansu handles all build uploads, signing, internal test tracks, beta distribution and App Store review responses.
2. Certificates, Identifiers & Profiles — Arial admin generates the App ID (e.g. com.arialengineering.aes), iOS Distribution certificate and App Store Distribution provisioning profile. We walk Arial's admin through every step in a single 30-minute screen-share call.
3. APNs Auth Key (.p8) — generated once from Arial's account, shared securely for push-notification setup. Revocable by Arial at any time.
4. App Store listing assets — Arial provides the final app icon and store description; we deliver screenshots, privacy policy URL and support URL.

Google Play Console (\$25 one-time — USD) — Arial creates the account in Arial's legal name. Arial adds Hitansu Parichha as Release Manager for the build and submission period. Hitansu manages all Play Store submissions, signing, testing tracks and review responses on both platforms throughout the build and warranty.

8.6 Package recommendation & flexibility on pricing

Recommendation: Professional Package (AES-Q-2026-003) at ₹6,79,500.

This is the only package that covers Arial's stated Phase 1 requirements — WhatsApp integration with full in-chat flow, Razorpay payment gateway, full AMC contract management, live engineer tracking, SLA escalation engine and 6 distinct user roles — without any add-ons.

On price flexibility

The headline contract value of ₹6,79,500 is firm and non-negotiable. This reflects the real, direct costs of delivering an enterprise-grade engagement — licensed code-security tooling, professional IDE and developer subscriptions, paid CI/CD minutes, monitoring platforms, design tools and secrets-management infrastructure

used to protect Arial's source code, credentials and customer data throughout the build. Reducing the price means removing these safeguards, which the Service Provider is not willing to do for a platform serving Arial's HNI/UHNI customer base.

Value-add concessions offered:

#	Concession	Standard Professional	What Arial gets
1	Revised 5-milestone payment schedule	30/20/30/20 across 4 milestones — Day 1 cheque: ₹2,03,850	20/20/20/20/20 across 5 equal milestones — Day 1 cheque: ₹1,35,900 (~₹68,000 less on signing day, ~33% softer upfront)
2	8 free training sessions across 4 months	3 sessions x 1 hour during go-live week	8 sessions x 1 hour across Months 1–4 post-go-live — covers all 6 roles, refreshers and open Q&A (effectively a 5-session upgrade worth ~₹40,000)

Both concessions are entered into the signed SOW v3 as Annexure B upon Arial's confirmation. The cost savings achieved by switching from cloud to dedicated server (~₹1,00,000+/year of avoided cloud bills) are real money in Arial's pocket from Year 1.

Section 9 — Supporting Documents & References

Materials available alongside this completed questionnaire

9.1 Which documents can you share alongside this questionnaire?

- ☒ High-level system / architecture diagram (updated for server-based hosting)
- ☒ API documentation / Postman collection
- ☒ Database schema overview (Flyway migration scripts)
- ☒ Security overview document (server-based, covers Section 5)
- ☒ Test plan / QA approach
- ☒ Portfolio — similar completed projects
- ☒ Cost Options Comparison Reference (AES_Cost_Options_Comparison.docx)
- ☒ Deep cost research write-up (COST_RESEARCH_DEEP_DIVE.md)
- ☐ Data flow diagram
- ☐ UI/UX prototype link (will be live in Staging during UAT week)
- ☐ Client references (contact details — shared on confidential follow-up call)
- ☐ None available at this stage

Not Available right now will send after phase 1

9.2 Client references — 1–2 previous clients Arial Engineering may contact:

Client references will be shared on a confidential follow-up call with Arial's leadership team rather than in this document, to protect prior clients' privacy. We are happy to arrange a separate reference call at a mutually convenient time.

9.3 Additional comments, commitments and context on record with Arial Engineering:

- Arial Engineering Services owns 100% of the source code, designs, data and intellectual property from the moment the final milestone payment is received.
- Written milestone schedule with minimum 5 business days advance notice on any expected delay; any delay is resolved at no additional cost to Arial.
- All customer data stays on Arial's dedicated server inside India. No third-party cloud storage is used for customer data.
- Ready to sign a mutual NDA, a data-processing agreement and a confidentiality addendum to the SOW v3 before project commencement.
- The development team is available for an in-person kick-off meeting in Hyderabad within the first week of the project.
- Phase 2 enhancements (multi-language, inventory management, accounting integration, IoT telemetry) scoped at a 10% partner-loyalty discount over standard rate card.
- Server runbook, Docker Compose configuration, Flyway migrations and full source code handover at Go-Live make Arial fully self-sufficient. Arial is never locked into the Service Provider for ongoing maintenance.

SIGN-OFF & AUTHORISATION

VENDOR TEAM	CLIENT — ARIAL ENGINEERING SERVICES
Full Name: Mahender Vengala	Full Name: Pradeep Reddy
Designation: Project Lead, Arial Engineering Solution	Designation: Arial Engineering Services
Date: 9 June 2026	Date: _____
Signature: Mahender Vengala	Signature: _____

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